

VentureIQ — Project Summary & Specification

AI Multi-Agent Startup Strategy & Validation Platform

1. Executive Overview

VentureIQ is an intelligent, multi-agent startup validation platform designed to analyze, stress-test, and refine startup concepts before founders commit capital or write code. Combining conversational AI discovery with a distributed multi-agent graph architecture (built on LangGraph and Gemini), VentureIQ evaluates market demand, competitive moat, business model viability, and execution risks in seconds.

2. Application Feature Matrix

Feature Area	Capability Specification & Implementation Details
Conversational Supervisor	Engages in dynamic follow-up questioning to extract target customer details, pricing models, and core value propositions before triggering deep analysis.
3D Intelligence Core	Interactive Three.js icosahedron core with glowing satellites and state-driven animations (<i>idle, thinking, question, analyzing, complete</i>).
Multi-Agent Pipeline	Executes 4 specialized analysis agents: Market (TAM/growth), Competitor (moat/landscape), Business (margins/CAC), and Risk (execution/regulatory).
Investor Readiness Index	Generates quantitative scorecards (0-100) across 4 pillars, classifying ventures into <i>Strong Venture Prospect</i> , <i>Viable Opportunity</i> , or <i>High Risk Profile</i> .
Session & Local History	Persists analyses locally with relative timestamp grouping (<i>Today, Yesterday, Last 7 Days, Older</i>). Allows instant switching between past briefs.
Multi-Format Brief Export	Supports <i>Copy Executive Brief</i> to clipboard, Markdown report export (.md download), and print-ready styled PDF view.

3. Methodology & Execution Flow

VentureIQ employs a four-phase state machine workflow combining conversational refinement with deterministic multi-agent graph execution:

- **Phase 1: Discovery & Prompting** — Founder submits initial idea via clean landing input or suggestion chips.
- **Phase 2: Supervisor Clarification Loop** — Lead Supervisor agent inspects context completeness. If crucial parameters are missing, it asks 1-3 targeted questions and offers interactive choice chips.
- **Phase 3: Multi-Agent Parallel Graph Execution** — Once readiness is confirmed, LangGraph orchestrates concurrent execution of Market, Competitor, Business, and Risk analysis agents.

- **Phase 4: Synthesis & Score Indexing** — Executive report generator consolidates individual agent outputs, calculates weighted Investor Readiness scores, and returns structured findings to the UI.

4. Technical Architecture

Layer	Technologies & Components
Frontend UI/UX	React 19, Vite 7, Tailwind CSS 4, Lucide Icons, Framer Motion. 3D Engine: Three.js r185, @react-three/fiber, @react-three/drei.
Backend API	FastAPI (Python 3.10), Uvicorn server, Pydantic data validation. Concurrency: In-flight deduplication locks (`IN_FLIGHT_LOCK`) to prevent race conditions.
AI Orchestration	LangChain (`langchain-google-genai`), LangGraph state workflows, Google Gemini 2.5/3.6 models with structured JSON parsing.
Persistence & State	Client-side localStorage history store (`ventureiq_conversations`), in-memory session cache (`SESSIONS` dict) on backend.

System Communication & Workflow Architecture

